Understanding the Value of a Firm

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Lesson 1: Goals of a Firm

Module : Understanding the Value of a Firm

Introduction

In this lesson we will cover the following topics:

- Different forms of corporations
- Goals of a firm and the role of management
- Introduction to firm value and opportunity cost of capital
- Investment and financing decisions
- Agency problems and corporate governance
- Summary and concluding remarks

Organization of a Corporation

Organization of a Corporation

A corporation, a firm, or a company is a legal entity that is owned by its shareholders

- In the eyes of the law, it is a legal person that can make contracts
- Corporations are formed under law, based on their articles of incorporation
- A corporation is owned by its shareholders, it is legally a different entity
- Often these shareholders have limited liability
- Shareholders cannot be held personally responsible for the corporation's debt

Organization of a Corporation

A corporation, a firm, or a company is a legal entity that is owned by its shareholders

- Often corporations start as proprietary (or partnership) firms
- Here, the owners (or partners) are solely responsible for the firm's actions with unlimited liability
- In these cases, the shares of the firm are not traded in financial markets
- Most large companies like Amazon, Apple, Meta (Facebook) are listed public companies with widely dispersed shareholdings

Organization of a Corporation

Separation of ownership and control gives permanence to corporations

- Even if managers are changed and shareholders sell shares, corporations survive
- However, this separation of ownership and control gives rise to managerial incentives that are not aligned to the owners
- This is called principal agent problem

Organization of a Corporation: Typical Balance Sheet

Assets		$ Billions	Liabilities		$ Billions
Current assets[a]		$2,037	Current liabilities[a]		$1,578
Fixed assets	$2,749		Long-term debt	$1,385	
Less depreciation	1,459		Other long-term liabilities[b]	1,105	
Net fixed assets		1,291	Total long-term liabilities[b]		2,490
Other long-term assets		3,515	Stockholder's equity		2,775
Total assets		$6,843	Total liabilities and stockholders' equity		$6,843

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Organization of a Corporation: : Typical Balance Sheet

	2008	2007
Assets		
Current assets:		
Cash and marketable securities	661	530
Accounts receivable	166	247
Inventories	8,209	7,611
Other current assets	215	298
Total current assets	9,251	8,686
Fixed assets:		
Tangible fixed assets		
Property, plant, and equipment	31,477	28,836
Less accumulated depreciation	8,755	7,475
Net tangible fixed assets	22,722	21,361
Long-term investments	253	509
Other long-term assets	460	313
Total assets	32,686	30,869

	2008	2007
Liabilities and Shareholders' Equity		
Current liabilities:		
Debt due for repayment	1,021	1,104
Accounts payable	4,543	4,137
Other current liabilities	2,458	2,510
Total current liabilities	8,022	7,751
Long-term debt	5,039	5,576
Deferred income taxes	660	670
Other long-term liabilities	910	774
Total liabilities	14,631	14,771
Common stock and other paid-in capital	735	729
Retained earnings and capital surplus	17,320	15,369
Total shareholders' equity	18,055	16,098
Total liabilities and shareholders' equity	32,686	30,869

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions. Chapter 28

Organization of a Corporation: Profit & Loss Statement

	$ millions
Net sales	48,230
Cost of goods sold	31,729
Selling, general, and administrative expenses	11,158
Depreciation	1,539
Earnings before interest and taxes (EBIT)	3,804
Interest expense	298
Taxable income	3,506
Tax	1,311
Net income	2,195
Dividends	491
Addition to retained earnings	1,704

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Introduction to Firm Value and Opportunity Cost of Capital

Introduction to Firm Value and Opportunity Cost of Capital

Maximization of firm value to shareholders is the ultimate goal of the management

- No two investors are the same in their age, taste, wealth, time-horizon, risk-tolerance, and investment strategy
- However, maximization of firm value is a common objective shared by all the shareholders
- It is up to shareholders whatever purposes they want to put this wealth
- They can do so even more if the same shares are worth more
- This maximization of firm value is indeed an optimum target if well-functioning financial markets are available

Introduction to Firm Value and Opportunity Cost of Capital

Financial markets facilitate investors to optimize the investment consumption pattern

- Financial markets also facilitate sharing of risk and risk diversification
- The goal of firm value maximization is different from maximizing profits, cash flows, or revenues
- Unlike firm-value, these performance measures can be manipulated to present a rosy picture
- Firm-value or share price is a more comprehensive measure of performance
- Thus, maximization of firm value is the most preferred and suitable objective for all the shareholders, irrespective of their tastes and preferences

Investment and Financing Decisions

Investment and Financing Decisions

To carry on business activity, a firm needs to invest in assets

- These assets generate long-term cash flows and the firm creates charge on these cash flows
- These financial securities are issued to investors to precisely fund these assets
- These financial securities are backed by the promise from the firm to repay the principal and interest on the loan
- This leads us to two key decisions made by firm: (a) Investment decision, and (b) Financing decision

Investment and Financing Decisions

The firm makes investment and financing decisions to smoothly run the firm operations

- Examples of Investment decisions include (a) Investment in plant and machinery, (b) Acquisition of mines, (c) Investment in research and development of products, (d) Investment into firm infrastructure
- These decisions are well planned in advance and often referred to as capital expenditures
- Most of these investments have returns (inflows) in the distant past

Investment and Financing Decisions

The firm makes investment and financing decisions to smoothly run the firm operations

- Examples of financing decisions include (a) Issuance of debt and equity securities, (b) Repayment of interest and principal obligations
- Financing decisions help organizations in raising money from lenders and shareholders
- This choice between debt and equity financing is termed as capital structure decision
- Capital is often referred to as long-term source of financing

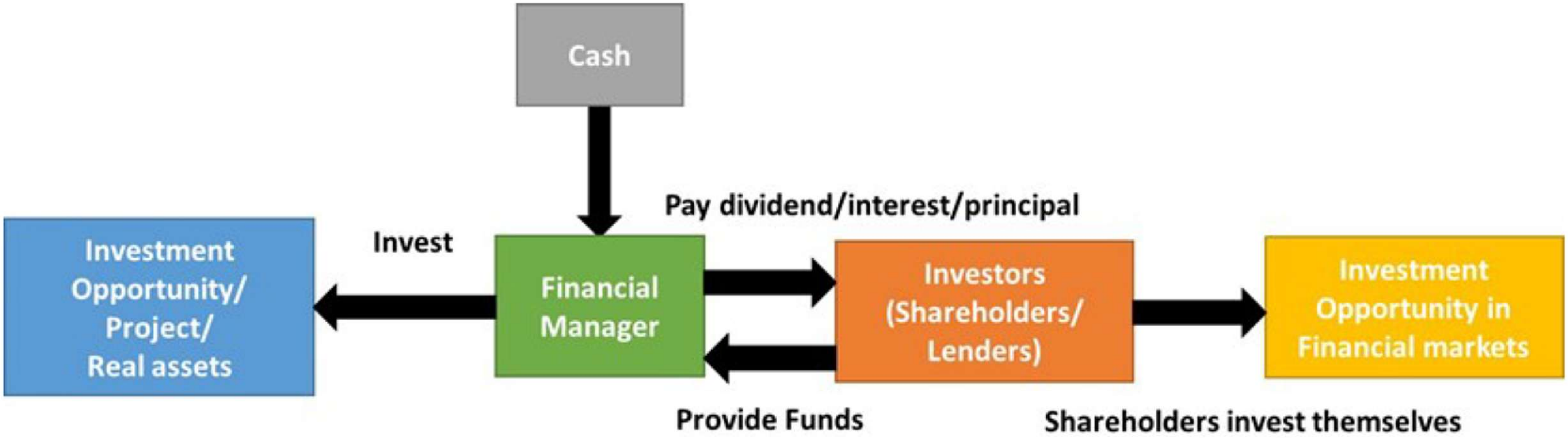
Investment and Financing Decisions

It is often debated amongst the financial managers that value comes from the investment side or financing side

- Investment side is termed as the asset side of the balance sheet and financing side is often termed as the liability side of the balance sheet
- If a firm A has a valuation of USD 1 Bn, what does it mean
- This is the current market value of the assets owned by the firm, due to the future earning potential of these assets

Investment and Financing Decisions

A simple illustration of investment and financing decisions is provided here



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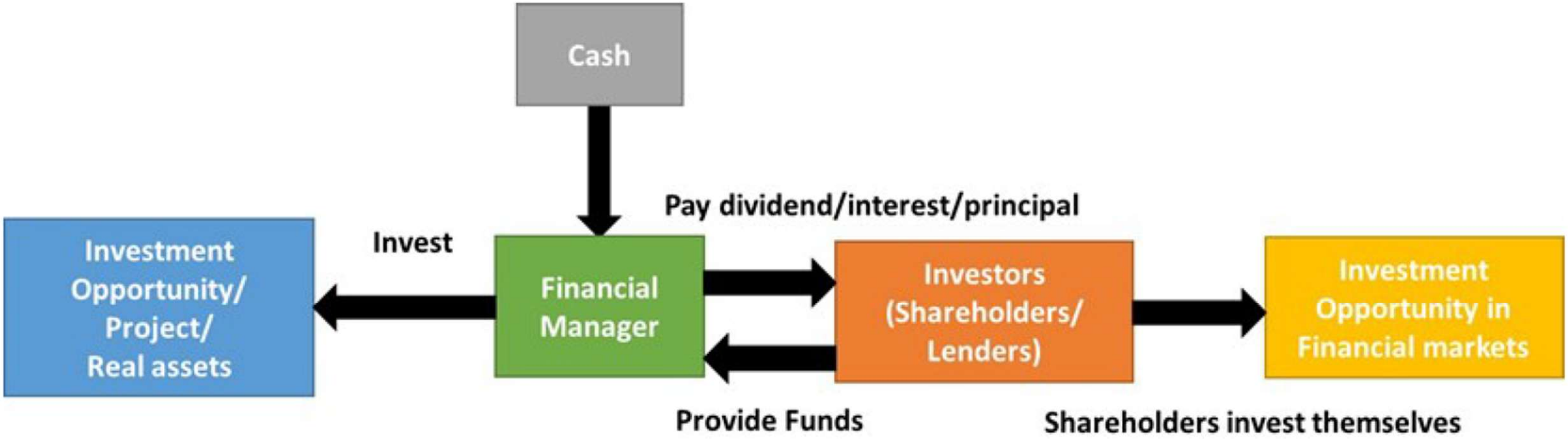
Investment and Financing Decisions

The simple answer is that if the rate of return on the project investment is higher than that available in the financial markets, then shareholders would want managers to invest in the project



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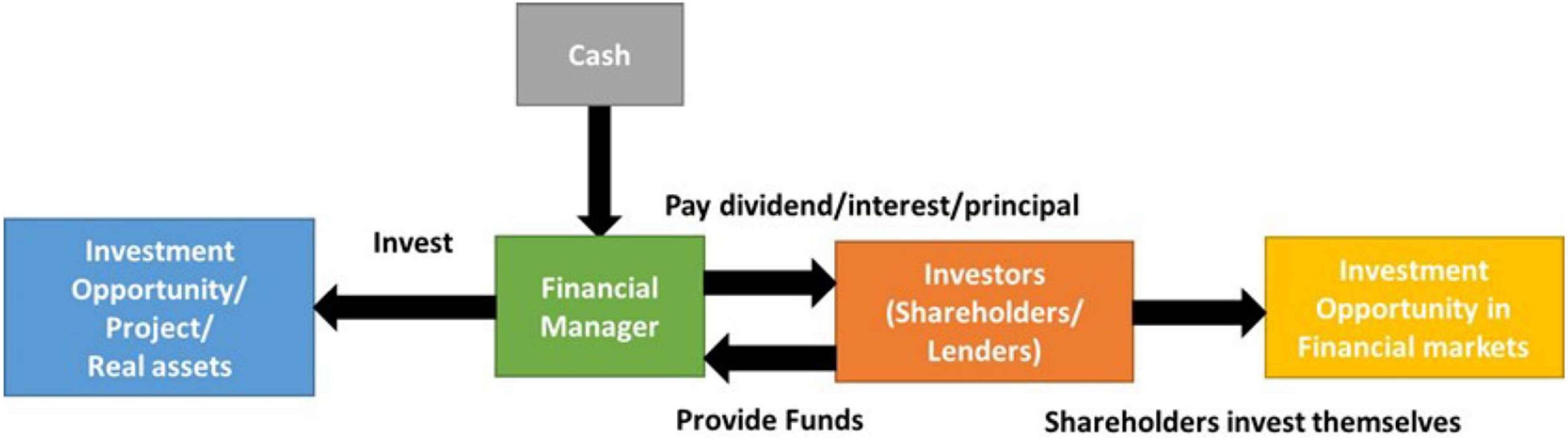
Investment and Financing Decisions

Any project investment is comparable to any financial market investment?



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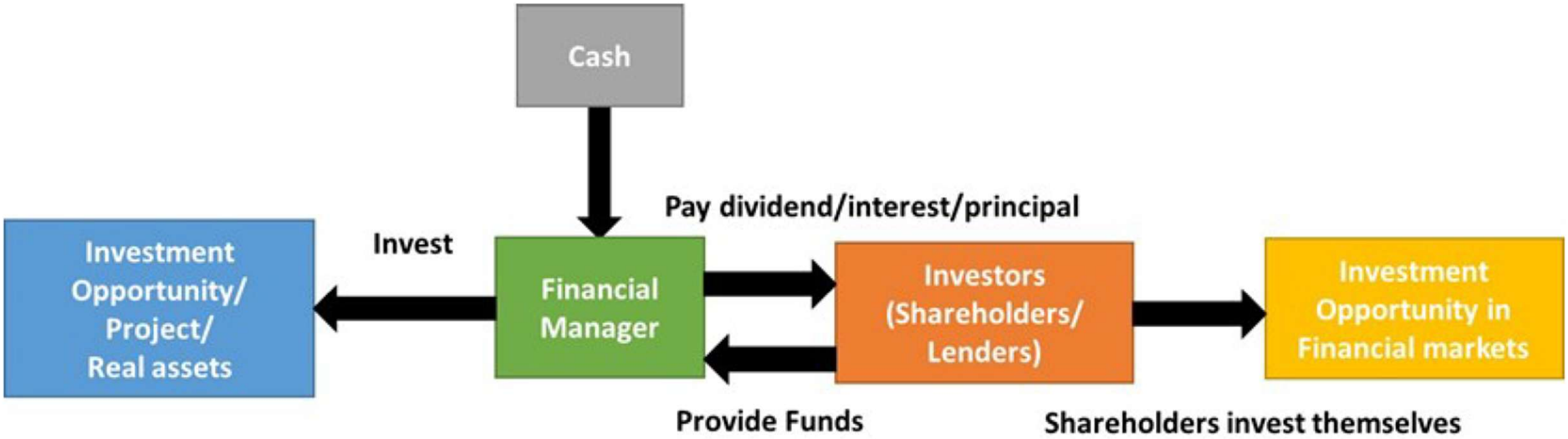
Investment and Financing Decisions

The return that is available in financial markets on similar risk instruments acts as the hurdle rate or cost of capital or “opportunity cost of capital”



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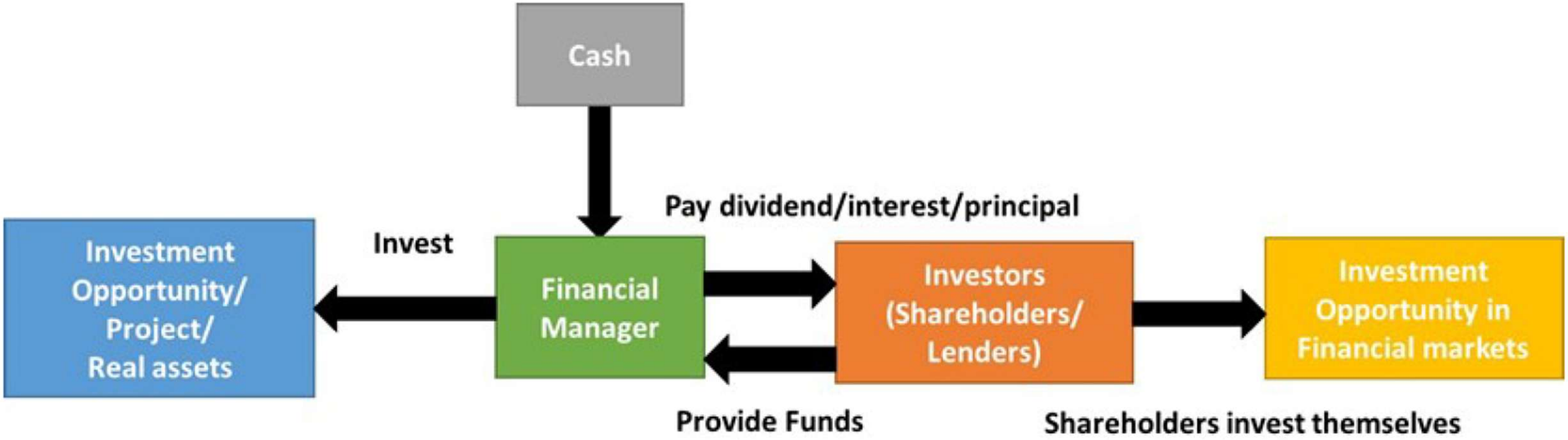
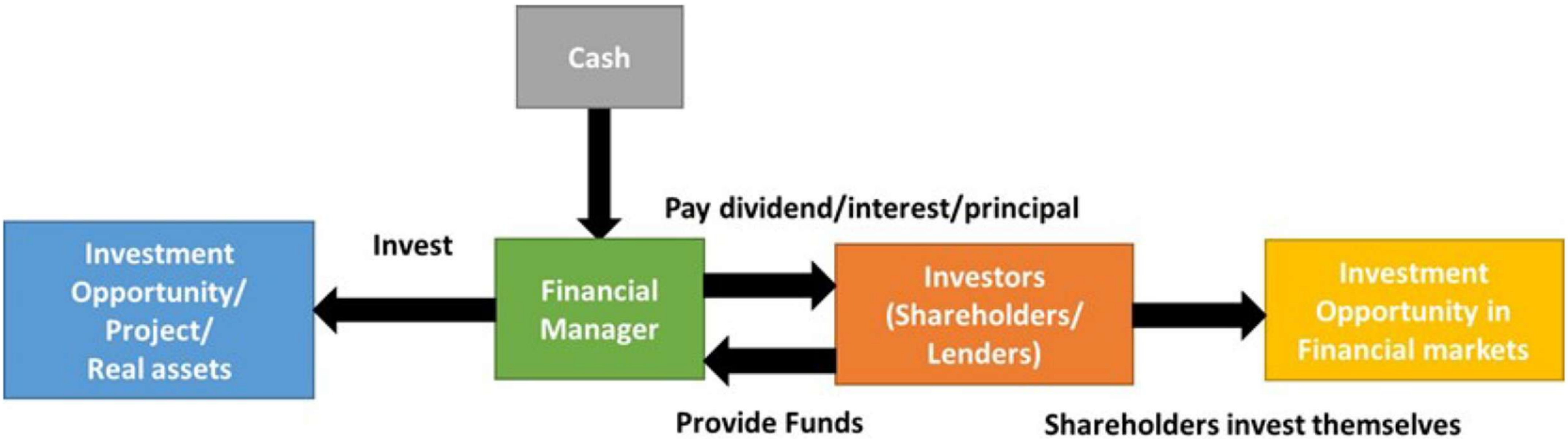
Investment and Financing Decisions

Corporations increase value by accepting all investment projects that earn more than the opportunity cost of capital



Investment and Financing Decisions

Investors are risk-averse. To hold any instrument of higher or lower risk, they would demand a higher or lower return according to their risk-averse nature



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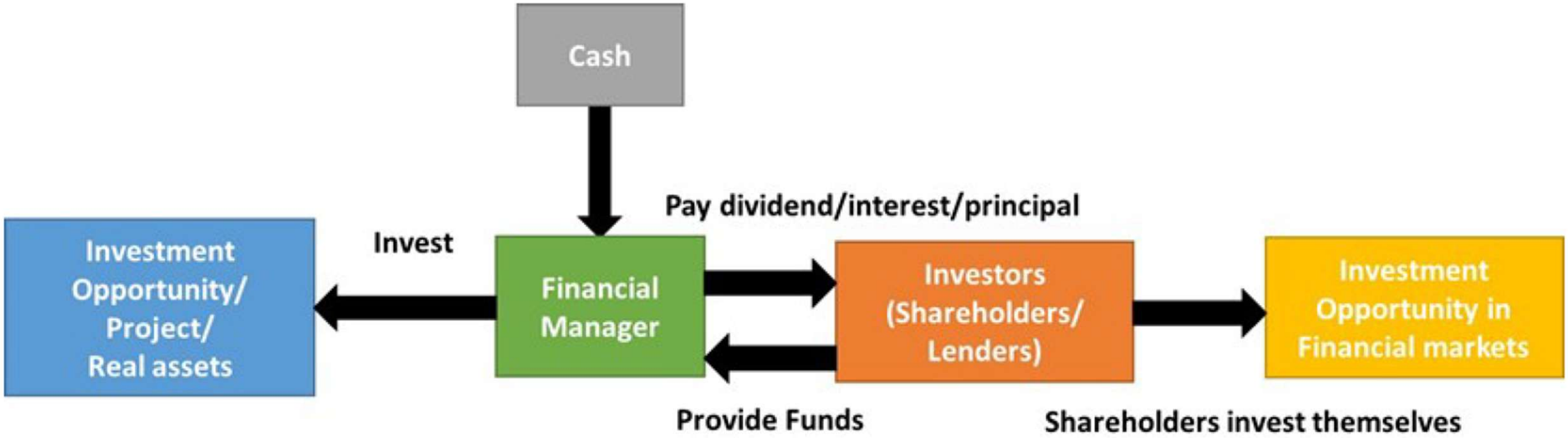
Investment and Financing Decisions

The fundamental assumption here is that financial instruments that are efficiently priced are available with different risk-return profiles



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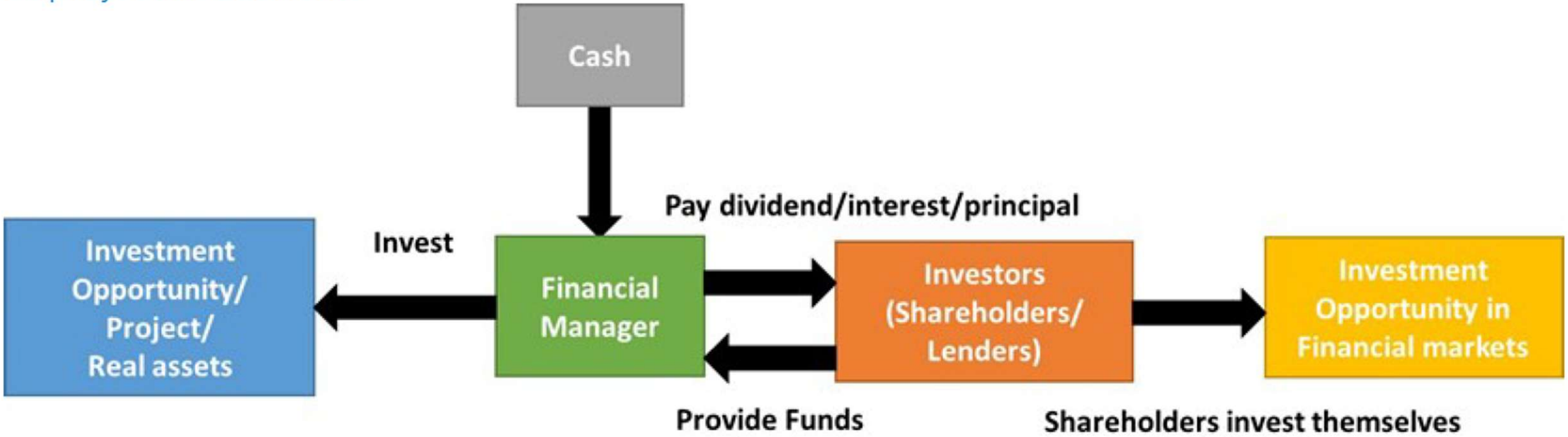
Investment and Financing Decisions

Investors look towards financial markets to measure the appropriate opportunity cost of capital for its project investments



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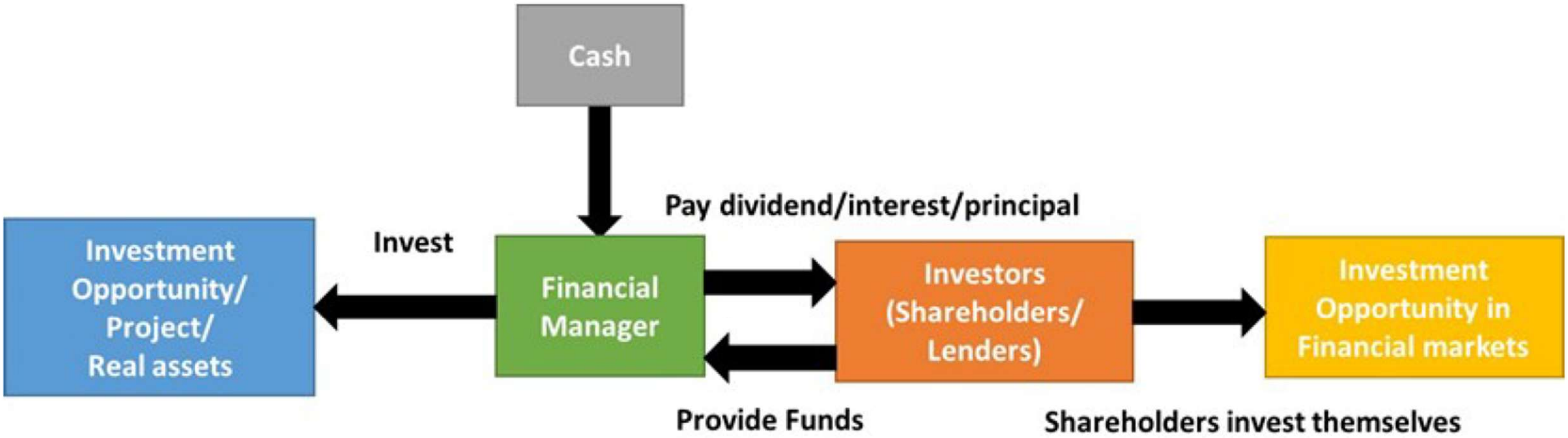
Investment and Financing Decisions

In general, projects are risky, and the appropriate interest rates are well above those that are available on risk-free government securities



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Agency Problems and Corporate Governance

Agency Problems and Corporate Governance

Shareholders want financial managers to maximize firm value as their ultimate goal

- For example, in order to increase future profits, managers may cut the current dividends and invest in projects with poor returns
- Managers are acting as the agents of their principals
- When managers act in the best interest of the firm, with firm value maximization being their ultimate goal, they also act in the best interest of other stakeholders such as customers, employees, partners, and their operating environment

Agency Problems and Corporate Governance

The goal of profit maximization is often misunderstood and considered as in conflict with the interest of other stakeholders

- Satisfied customers, loyal employees, and a harmonious operating environment enhance the firm value
- Shareholders do not directly control the day to day operations of the managers
- They only control the long-term strategic actions of managers through the institution of the board of directors
- Shareholders can show their approval or disapproval by selling and buying the shares

Agency Problems and Corporate Governance

Separation of ownership and control though necessary, but also has some negative consequences

- Managers may take certain actions that may not be in the best interest of shareholders
- They may avoid projects with long-term value creation, in favor of the projects with short-term benefits
- Conflicts between the managers and owners are called agency problems
- Agency problems arise, when agents work for principals
- Shareholders are the principals and managers are their agents
- Agency costs are incurred when managers do not aim to maximize the firm value

Agency Problems and Corporate Governance

In order to mitigate these agency costs good systems of corporate governance are needed

- The design of compensation schemes for the top management
- Legal and regulatory requirements related to accounting and reporting standards for listed firms to ensure consistency and transparency
- Board-of-director institution and monitoring by security analysts
- Shareholder pressure through buying and selling of stocks
- Market discipline through threat of takeovers
- These mechanisms help align the interests of managers to those of shareholders

Summary and Concluding Remarks

Summary and Concluding Remarks

- Two important financial decisions: investment decision and financing decision
- Value maximization as the ultimate objective of managers
- Trade-off faced by financial managers and opportunity cost of capital
- Separation of ownership and control
- Principal agent problem
- Good systems of corporate governance

Thanks!